LeetCode 66 – Plus One

1. Problem Title & Link

Title: LeetCode 66: Plus One

Link: https://leetcode.com/problems/plus-one/

2. Problem Statement (Short Summary)

You are given a large integer represented as an array of digits.

Each element contains one digit.

Add **1** to the number and return the resulting array.

Example

Input:

digits = [1,2,3]

Represents:

123

After adding one:

124

Output:

[1,2,4]

3. Examples (Input → Output)

Example 1

Input:

digits = [1,2,3]

Output:

[1,2,4]

Example 2

Input:

digits = [4,3,2,1]

Output:

[4,3,2,2]

Example 3

Input:

digits = [9]

Output:

[1,0]

Example 4

Input:

digits = [9,9,9]

Output:

[1,0,0,0]

4. Constraints

1 <= digits.length <= 100

0 <= digits[i] <= 9

Important:

No leading zeros.

Except:

0 itself

5. Core Concept (Pattern / Topic)

Array Manipulation

Secondary Topics:

- Carry Propagation
- Digit Simulation

6. Thought Process (Step-by-Step Explanation)

Normal Case

Example:

123

Adding one:

123 + 1

124

Simply increment last digit.

Problem Case

Example:

129

129 + 1

130

Why?

9 + 1 = 10

Carry generated.

Bigger Example

999 + 1

Step 1

999

↑

9+1=10

Store:

0

Carry:

1

Step 2

990

↑

9+1=10

Store:

0

Carry remains.

Step 3

900

↑

9+1=10

Store:

0

Carry remains.

Final:

1000

Need new digit.

Key Observation

Process digits from:

Right → Left

because addition starts from the least significant digit.

7. Visual / Intuition Diagram (ASCII Diagram)

```
Example:
[1,2,9]
      1
      ↓

[1,2,9]

     ↑

9+1=10

[1,2,0]

Carry=1

Next:
2+1=3
Result:
[1,3,0]

Example:
[9,9,9]
9 → 0

9 → 0

9 → 0

Need new digit
```

```
↓

[1,0,0,0]
```

8. Pseudocode (Language Independent)

```
for i from n-1 to 0

    if digit < 9

        digit++

        return digits

    digit = 0

create new array

newArray[0] = 1

return newArray
```

9. Code Implementation

Python

```python
class Solution:
    def plusOne(self, digits):

        n = len(digits)

        for i in range(n - 1, -1, -1):

            if digits[i] < 9:
                digits[i] += 1
                return digits

            digits[i] = 0

        return [1] + digits
```

Java

```java
class Solution {
```

```
    public int[] plusOne(int[] digits) {

        for(int i = digits.length - 1; i >= 0; i--) {

            if(digits[i] < 9) {

                digits[i]++;
                return digits;
            }

            digits[i] = 0;
        }

        int[] result = new int[digits.length + 1];

        result[0] = 1;

        return result;
    }
}
```

10. Time & Space Complexity

Time Complexity

O(n)

Worst Case:

999999999

Need to traverse all digits.

Space Complexity

O(1)

Normal case.

O(n)

Only when new array needed.

11. Common Mistakes / Edge Cases

Mistake 1

Only incrementing last digit.

Fails:

[1,2,9]

Mistake 2

Ignoring carry.

Fails:

[9,9]

Mistake 3

Not creating larger array.

Fails:

[9,9,9]

Expected:

[1,0,0,0]

12. Detailed Dry Run

Input:

[9,9,9]

Index	Value	Action
2	9	0
1	9	0
0	9	0

Array becomes:

[0,0,0]

Need extra digit:

[1,0,0,0]

Output:

[1,0,0,0]

13. Common Use Cases (Real-Life / Interview)

Big Integer Arithmetic

Languages sometimes store huge numbers as arrays.

Example:

99999999999999999999

Cannot fit into normal integer.

Calculator Software

Digit-by-digit addition.

Banking Systems

Large precision arithmetic.

14. Common Traps (Important!)

Trap 1

Converting array to integer.

Example:

[9,9,9,9,9,9,9]

Works here but not for huge numbers.

Interviewers dislike this solution.

Trap 2

Forgetting:

[9]

becomes:

[1,0]

Trap 3

Not returning immediately after successful increment.

15. Builds To (Related LeetCode Problems)

LC 66 - Plus One

↓

LC 989 - Add to Array Form

↓

LC 415 - Add Strings

↓

LC 2 - Add Two Numbers

Excellent progression for digit manipulation.

16. Alternate Approaches + Comparison

Approach	Time	Space	Interview Rating
Convert to Integer	O(n)	O(n)	Poor
Carry Simulation	O(n)	O(1)	Best

17. Why This Solution Works (Short Intuition)

Addition always starts from the rightmost digit.

Whenever a digit becomes:

10

we store:

0

and carry:

1

to the next digit.

If every digit is 9, we create a new leading 1.

18. Variations / Follow-Up Questions

Follow-Up 1

Add K instead of 1.

Related:

LC 989

Follow-Up 2

Subtract One.

Example:

[1,0,0]

↓

[9,9]

Follow-Up 3

Add Two Large Numbers stored as arrays.

Follow-Up 4

Multiply two large numbers stored as arrays.